

Pharmacokinetics Worksheet (Examples)

1. If a dose of 500 mg of a drug result in a plasma concentration of 25 mg/L, what will be the volume of distribution the drug?

Ans. 20L

2. Calculate the half-life of a drug, assuming a first order and one compartment disposition, when 20% of the dose remains in the body 6 hours after an IV bolus dose.

Ans. 2.6 hrs.

3. Assume the volume of distribution is 20 L and the clearance is 5 L/hr, what will be the half-life this drug?

2.77 hrs.

4. The loading dose is the initial high dose of a drug given to rapidly achieve a therapeutic concentration. A Physician intends a desired plasma concentration of 25 mg/L, if the volume of distribution of the drug is 20 L, what loading dose will provide the desired concentration?

500 mg

5. Estimate the volume of distribution of diclofenac sodium 100mg given IV as a bolus dose. The plasma concentration was determined to be 2.5 mg/L just after 12hrs the dose is given. (Elimination half-life of the drug is 12h)

Ans. 20L

6. A syringe pump is delivering voriconazole infusion 200 mg/50 mL at a rate of 2.5 mL/hour to a male patient with a body weight of 75 kg. calculate the dose in µg/kg per hour that the patient is receiving?

Ans. 133

7. A patient receives dose of a drug D, and plasma concentrations were 22 and 15 mg/L at 24 and 48 hours after infusion, respectively. Estimate the elimination half-life of the drug?
8. When 100 mg of a drug was given as an IV bolus dose, the following plasma concentration time relationship (C in mg/L & t in hr.) was observed:

$$C = 7.14e^{-0.173t}$$

Calculate:

- a) Volume of distribution
 - b) The elimination half life
 - c) Systemic clearance
 - d) AUC
 - e) The plasma concentration 20min. after a 250 mg IV bolus dose.
9. Theophylline is administered to a patient at 30 mg/hour via a constant IV infusion. If the patient has a total body clearance for theophylline of 50 mL/minute, what should this patient's steady-state plasma concentration be?

Ans. 10.0 mg/L

10. Patient D receives drug X, 100 mg every 8 h IV. At steady state, two plasma concentrations are measured: 9.6 mg/L (1h post dose) and 2.9 mg/L (2h pre-dosing).
- a. Calculate the elimination rate constant (K) (Ans. $K = 0.171/h$)
 - b. Calculate the elimination half-life ($t_{1/2}$) (Ans.)
 - c. Calculate the volume of distribution (V_d) (Ans. 11.8L)
11. The data in the following table is obtained on infusing a drug at a rate of 50 mg/hr. for 7.5 hrs.

Time (hrs.)	0	2	4	6	7.5	9	12	15
Cp (mg/L)	0	3.4	5.4	6.5	7	4.6	2	0.9

Calculate:

- a) Volume of distribution
 - b) Elimination rate constant
 - c) Elimination half-life
 - d) Systemic clearance
12. The population pharmacokinetics of a given drug for a 70 kg person are: $F = 0.65$, $V = 280L$, and $CLs = 4.5 L/hr$. Given that absorption is instantaneous relative to elimination, calculate the following when a 50mg dose of the drug is taken daily:
- a) The maximum and minimum concentration of the drug in the boy at steady state
 - b) The accumulation ratio
 - c) The time require to reach to 50% of the steady state concentration.

13. A patient with liver failure and a patient with heart failure need to be treated with a new antiarrhythmic drug. You find a research study that contains the following information for the drug in patients similar to the ones you need to treat: normal subjects: clearance = 45 L/h, volume of distribution = 175 L; liver failure: clearance = 15 L/h, volume of distribution = 300 L; heart failure: clearance = 30 L/h, volume of distribution = 100 L. Recommend an intravenous loading dose and continuous intravenous infusion maintenance dose (infusion rate) to achieve a steady-state concentration of 10 mg/L for your two patients based on this data and estimate the time it will take to achieve steady-state conditions.
14. Table below gives the plasma drug concentrations (C_p) that were obtained following oral administration of 1 g dose of a drug. Plot the data and, using the plot, determine the following.
- The elimination half-life ($t_{1/2}$) and the elimination rate constant (K).
 - The absorption rate constant (K_a).
 - The observed and calculated peak time (t_{max}).
 - The observed and calculated peak plasma concentrations, $(C_p)_{max}$.
 - The intercept

Time (h)	Plasma concentrations (mg)
0.25	3.00
0.50	4.60
1.00	5.70
1.50	5.60
2.00	4.80
3.00	3.20
4.00	2.00
5.00	1.20
6.00	0.75
7.00	0.46